Hiyd Fitness Tracker – Electrical Technical Design Specification (TDS)

Design and development of a compact, high-performance fitness tracker

Prepared by
Omar Shakir
Hiyd Engineering Lead

Date
May 2025 – V1.0

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1.0 Table of Contents

Contents

1.0 Table of Contents 2
2.0 Introduction 4
3.0 System Requirements 5
3.1 Overview 5
3.2 Power Management: 5
3.3 Data Collection and Processing: 5
3.4 Connectivity and Communication: 6
3.5 Functional Requirements: 6
3.6 Physical and Mechanical Requirements: 6
3.7 Durability and Safety: 7
3.8 Data Processing and Algorithm Requirements: 7
3.9 Communication & Integration: 7
4.0 Proposed Components 8
5.0 Firmware/Software Interface 9
5.1 MCU-Firmware Interface 9
5.2 BLE Protocol Stack 9
5.3 Companion App Interface 9
6.0 Testing and Validation 10
6.1 Initial Electrical Testing 10
6.2 Sensor Verification 10
6.3 Battery & Power Validation 10
6.4 BLE Communications & Range 10
6.5 Thermal and Environmental Testing 10
7.0 Production Plan 11
7.1 Phase 1: Prototype Validation (1–10 units) 11
7.2 Phase 2: Engineering Test Batch (100 units) 11
7.3 Phase 3: Initial Production (1000 units) 11

8.0 Version History 12

2.0 Introduction

This document outlines the electrical design specifications for the HIYD Fitness Tracker, a next-generation wearable device that integrates SpO2 monitoring, heart rate tracking, accelerometer-based motion sensing, and GPS functionality within a compact, high-performance form factor. The tracker is engineered to provide continuous, real-time insights into users' health metrics, empowering them to optimize their fitness routines and monitor vital parameters.

The electrical system design outlined in this document covers the hardware components, power management, sensor integration, and system architecture required to achieve optimal performance. A separate specification document will detail the companion mobile application, which will interact with the tracker to visualize and analyze user data, as well as provide additional features such as activity logging, goal setting, and notifications.

This document serves as a comprehensive guide for the electrical design, development, and testing of the HIYD Fitness Tracker, ensuring that all systems are integrated efficiently to deliver a robust and reliable wearable solution.



Figure 1: Overview of Hiyd tracker attaching to watch

3.0 System Requirements

3.1 Overview

- Form Factor for PCBA + Battery: Ø28mm x 1.7 mm thick.
- Battery must remain within the overall form factor outlined above.
- Heart Rate and SPO2 monitor must be placed in center of tracker, within lens to collect reflected light.

3.2 Power Management:

- Low Power Operation: System must be capable of operating efficiently, with battery life of 5+ days in passive mode, and approximately 1 day in active mode.
- Smart Power Saving: Ability to switch between sleep, low-power, and active modes based on activity (e.g., heart rate monitoring, accelerometer, GPS use).
- Battery Monitoring: Monitor battery status and indicate when charging is needed.
- Charging Interface: Wireless or contact pin charging capability.

3.3 Data Collection and Processing:

- Heart Rate Monitoring: Continuous measurement of heart rate with a pulse oximeter (SpO2) to monitor oxygen saturation.
- SpO2 Monitoring: Collect blood oxygen levels with the integrated pulse oximeter.
- Accelerometer Data: Detect and log motion, activity, and steps.
- Gyroscope Data: Measure and track rotational movements and orientation (useful for detecting specific activities or postures).
- Temperature Monitoring: Track body temperature continuously or on demand, with the possibility of alerting the user to abnormal body temperature readings or menstrual cycle monitoring.
- GPS Data (Optional): Provide geolocation tracking for outdoor activities (such as running or walking), with integration to a mobile app. If possible to integrate in form factor.
- Real-Time Clock (RTC): Maintain accurate timekeeping for timestamping sensor data and managing sleep modes.

3.4 Connectivity and Communication:

- Bluetooth (BLE 5.0): Wireless communication with cellphone for syncing data.
- Mobile App Integration: Sync collected data with a companion app.
- Over-the-air (OTA) firmware updates.
- Push notifications and alerts based on user-set thresholds (e.g., heart rate, SpO2 levels).

3.5 Functional Requirements:

- User Interface: No screen on device - data transferred to mobile app.
- Visual feedback through LED indicators (e.g., charging status, battery status).
- Data Logging: Ability to store data for 1 day on device before requiring sync with app.
- Sensor Calibration: Automatic calibration of sensors (heart rate, SpO2, accelerometer).
- Error Handling: Robust error detection and handling for all sensors.
- Notifications for abnormal sensor readings (e.g., error in heart rate measurement).

3.6 Physical and Mechanical Requirements:

- Form Factor for PCBA + Battery: Ø28mm x 1.7 mm thick (excludes enclosure and lens). Refer to figures 2 and 3 below for more details.
- Water Resistance: IP67 or IP68 to withstand sweat, splashes, and brief submersion.
- Shock Resistance: Ensure PCBA is resilient to light impacts and normal wear and tear.

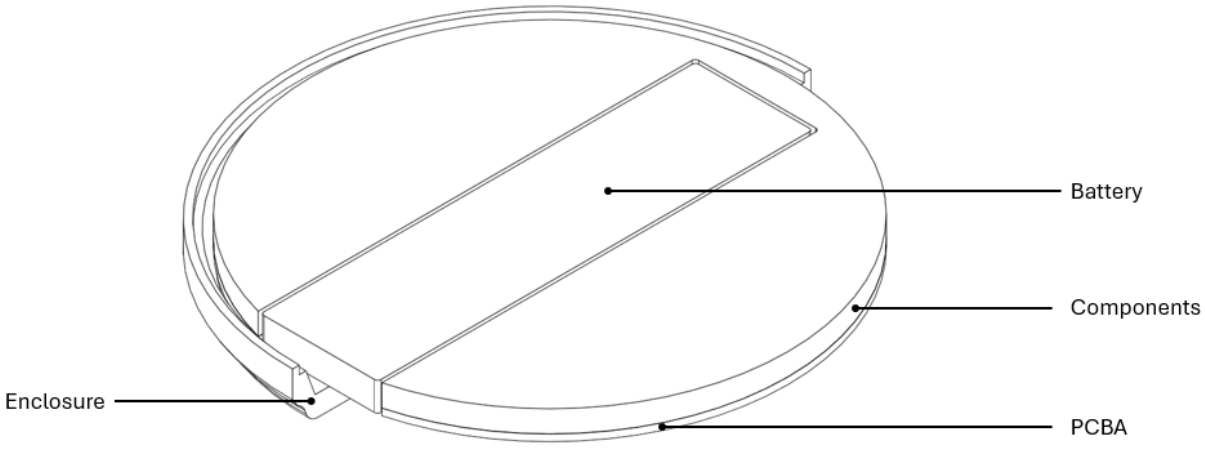


Figure 2: Sample Tracker Section View (Isometric)

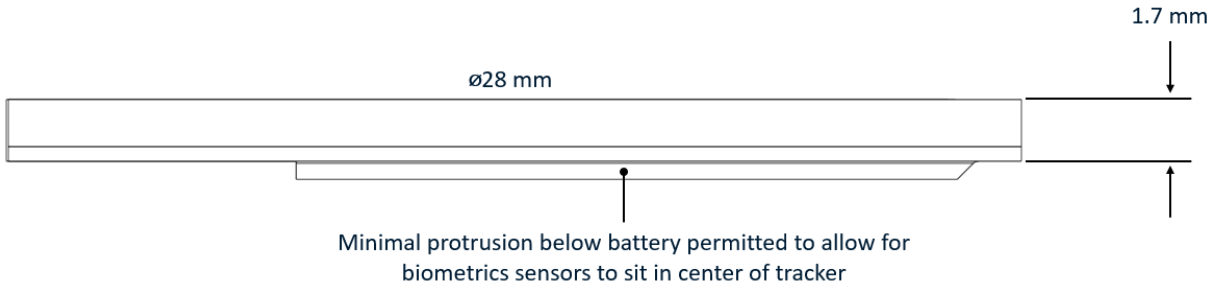


Figure 3: Sample Overview of Form Factor

3.7 Durability and Safety

- Battery Protection: Must include protection for overcharge, over-discharge, and short circuits to ensure safe battery usage.
- EMI Shielding: Must have electromagnetic shielding to prevent interference.
- Regulatory Compliance: CE, FCC, and RoHS regulations for safe use in various regions.

3.8 Data Processing and Algorithm Requirements:

- Sensor Fusion: Use algorithms to combine accelerometer, gyroscope, and heart rate data to detect specific activities or patterns.
- Step counting and activity classification (e.g., running, walking, cycling) using sensor fusion data.
- Health Insights: Provide users with health insights based on heart rate, SpO2, and temperature data (e.g., trends, abnormal readings, activity levels).
- GPS Tracking (if included): Provide real-time tracking during outdoor activities.
- Store GPS coordinates for route history and display them in the mobile app.

3.9 Communication & Integration:

- Bluetooth LE (BLE) 5.0: Seamless data transfer between the tracker and the companion mobile app.
- Low energy consumption, optimizing battery life while maintaining sync with the app.
- Mobile App: Integrate with both iOS and Android for user-friendly experience.
- Display real-time data, notifications, health metrics, and historical trends.
- Provide customization for user preferences (e.g., alert thresholds for heart rate or SpO2).
- Handle firmware updates over-the-air (OTA).

4.0 Proposed Components

Component	Part Number/Model	Size (mm)	Capacity/Specs	Notes
Battery	LiPol Battery - LPCV170726	26.00 x 7.00 x 1.70	18 mAh	-Fits within 1.7 mm thickness; will need to be optimized to permit reasonable usage -May need to size to larger battery -Typical fitness ring battery
Heart Rate + SpO2 Sensor	MAX30102	5.60 × 3.30 × 1.55	Dual LED (IR + Red)	Combines HR and SpO2 sensing in one module
Accelerometer	Bosch BMA456	2.00 × 2.00 × 0.65	3-axis accelerometer, low power mode	Detects motion, steps, and activity
Gyroscope	STMicro L3GD20	3.00 × 3.00 × 1.00	3-axis gyro	Tracks rotation and orientation
Temperature Sensor	ADT7420	3.00 × 3.50 × 0.90	Precision temperature sensor	Monitors body temperature (important for health monitoring)
Microcontroller (MCU)	Nordic nRF52832 (WLCSP)	3.00 × 3.20	BLE 5.0, Cortex-M4F, 512KB Flash	Controls all components, Bluetooth connectivity
Crystal (RTC)	32.768 kHz	1.60 × 1.00	Low power RTC	Real-time clock, ultra-low power consumption
Decoupling Capacitors	0402 / 0201	<1.00 × <1.00	0.1 µF, 0.01 µF, 10 µF	Filters power supply noise, stabilizes voltage
Resistors	0402 / 0201	<1.00 × <1.00	Various values, low power	Voltage dividers, biasing, and filtering components
Inductors	0402 / 0201	<1.00 × <1.00	Low power (optional, for RF/Filtering)	Optional for signal conditioning and power filtering
PCB	HDI Flex-Rigid	Ø 28.00 x 1.70 Cutout for battery	Multi-layer design (2-4 layers)	Semi-circular PCB, low-profile stackup, flexible options
PCB Connector (Battery)	Flexible PCB Connector	N/A	Small, flexible	Connects battery to PCB, allows for flexibility in design

5.0 Firmware/Software Interface

This section describes how the embedded firmware and companion application will interface with hardware components.

5.1 MCU-Firmware Interface

- Real-time event-driven firmware on the nRF52832 (or similar).
- Drivers for:
  - I^2C: MAX30102 (HR/SpO_2), BMA456 (accel), temperature.
  - SPI/UART (optional): GPS module.
  - Low-power sleep and wake states managed by firmware scheduler.

5.2 BLE Protocol Stack

- GATT-based BLE communication.
- Custom Services:
  - Sensor Data Service (HR, SpO_2, accel, temp, GPS).
  - Battery Status Service.
  - Device Info & OTA Updates.

5.3 Companion App Interface

- Connect via BLE and syncs data.
- Receives sensor payloads and displays analytics.
- Handle user settings, calibration prompts, firmware updates.

Note: Application development and interface logic will be handled in a separate technical specification document.

6.0 Testing and Validation

This section outlines the procedures required to ensure that the electrical system meets all functional, safety, and performance criteria prior to scaling production.

6.1 Initial Electrical Testing

- Continuity & Short Checks (post-assembly).
- Power-On Verification (voltage rails, current draw at boot).
- Functional Test of sensors, BLE, and MCU behavior on dev board.
- Oscilloscope & Logic Analyzer Use to verify signal integrity, I^2C/SPI/BLE transactions.

6.2 Sensor Verification

- Confirm HR & SpO_2 accuracy under different skin tones and conditions (light, sweat).
- Accelerometer: tap detection, motion patterns, orientation accuracy.
- Gyroscope: stability and drift testing.
- Temperature sensor: environmental and skin calibration check.
- GPS module: fix time, signal strength indoors vs outdoors.

6.3 Battery & Power Validation

- Confirm charging time, discharge rate under passive and active use cases.
- Validate 1-day active use and 5-day passive standby.
- Monitor for overcurrent/overvoltage protection functionality.

6.4 BLE Communications & Range

- Range testing in line-of-sight and obstructed environments.
- Throughput and latency testing for BLE notifications and sensor sync.

6.5 Thermal and Environmental Testing

- Perform testing at 0°C, 25°C, and 40°C to validate sensor consistency.
- Drop test and mild vibration simulation (if enclosure ready).
- Ship testing.
- Enclosure sealing testing (if enclosure is ready). IP67 rating preferred.

7.0 Production Plan

This section outlines the path from prototype to production.

7.1 Phase 1: Prototype Validation (1–10 units)

- Build and hand-test ~5–10 units.
- Validate electrical, firmware, and sensor functionality.
- Identify design or manufacturing issues before scaling.

7.2 Phase 2: Engineering Test Batch (100 units)

- Contract manufacturing with consistent assembly process.
- Full DVT (Design Validation Testing): performance, power, wireless, data integrity.
- Collect field data via limited beta users.
- Firmware refinements, BLE tuning, edge-case testing.

7.3 Phase 3: Initial Production (1000 units)

- Finalized electrical and mechanical design.
- Manufacturing SOP and test jig implementation.
- App store launch for companion application.
- Begin certification (FCC, CE, etc.) if required.
- Prepare for public or soft launch.

8.0 Version History

Version Number	Notes	Prepared By
1.0	Initial Release.	Omar Shakir